Short-term reliability of behavior in the zebrafish novel-tank test: a six-session reanalysis

Generated by A.I. (Claude Opus 4.8, Claude Sonnet 4.6, Gemini Pro 3.1)

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Abstract

The novel-tank diving test is the most widely used assay of anxiety-like behavior in adult zebrafish, with the proportion of time spent in the upper half of the tank (pct_top) as its primary index. For this measure to index stable individual differences, it must be repeatable — yet most zebrafish repeatability estimates come from intervals of weeks to months, and short-term, within-days reliability is largely unexamined. We reanalyzed a six-session dataset (103 adult zebrafish tested twice daily over three days) with a newly built Python pipeline that re-implements our laboratory's earlier Mathematica analysis from the same distortion-corrected tracking. The new pipeline reproduced the original internal consistency (Cronbach's $\alpha = 0.855$ across the six sessions; original $\alpha \approx 0.88$). Expressed as the single-measure repeatability standard in the animal-personality literature, pct_top showed $R \approx 0.50$ per session; the composite α is simply the Spearman–Brown projection of that value over six sessions, and once placed on the same scale the estimate sits at the upper end of published (mostly long-interval) values. The first, novel exposure ranked individuals differently from all later sessions — repeatability rose to $R \approx 0.59$ when it was excluded — consistent with acclimation before behavior stabilizes; within-day and day-to-day reliability likewise strengthened after the first day. Among additional measures, lateral movement range and swimming velocity were as repeatable as pct_top, transition frequency less so, and latency to first ascent least. We draw two practical recommendations: report novel-tank reliability as a variance-partitioned single-measure R with the number of sessions stated, not as a composite α or a raw test–retest correlation; and treat the first exposure as an acclimation session distinct from familiar-tank measurement. Two limitations bound interpretation: a within-days window measures short-term repeatability of among-individual differences, which need not reflect a lifelong personality trait and may include persistent post-handling or physiological state; and the original frame-differencing tracker does not register freezing (interpolation recovers within-session bottom freezes, but immobility is only partially observed).

Keywords: zebrafish; novel-tank test; behavioral repeatability; short-term reliability; animal personality; intraclass correlation.

A.I.-generated work — read first. This manuscript and the analysis pipeline it describes were generated by A.I. (largely Claude Opus 4.8, Claude Sonnet 4.6, and Gemini Pro 3.1, with human orchestration). The underlying tracking data, derived from the original Mathematica code, are sound; nearly everything else — the Python code and this write-up — is A.I.-generated and should be treated as such. Readers are encouraged to re-derive any result of interest from scratch to verify it. If you find errors, please open a GitHub issue.

Status. This work was presented previously only as a conference poster and has **not** been peer-reviewed or published in a journal. It is a *within-laboratory* A.I.-driven reanalysis (a re-implementation of an earlier human-driven analysis), **not** an independent replication. Results come from a single cohort at one facility and are shared for transparency and reuse, not as settled findings.

1 Introduction

The adult zebrafish (*Danio rerio*) has become a standard model for the behavioral neuroscience of anxiety and stress (Stewart et al. 2012; Maximino et al. 2010). The central assay is the novel-tank diving test, in which a fish placed in an unfamiliar tank initially dives and remains near the bottom, then gradually explores the upper water column as it habituates. The proportion of time spent in the upper half — equivalently, reduced bottom-dwelling — is the field's primary anxiety index, validated by its sensitivity to anxiogenic and anxiolytic manipulations and its correspondence with whole-body cortisol (Egan et al. 2009; Cachat et al. 2010; Levin, Bencan, and Cerutti 2007). The test is inexpensive, high-throughput, and amenable to automated video-tracking, which has made it a workhorse for pharmacological and genetic screening (Tran and Gerlai 2016; Stewart et al. 2012).

If top-half occupancy is to be interpreted as an index of an individual's *trait* anxiety rather than a momentary state, it must be repeatable: fish that are relatively bold on one occasion should remain relatively bold on another. This is the defining requirement of the animal-personality framework, in which repeatability — the proportion of total behavioral variance attributable to stable among-individual differences — sets an upper bound on how much a measure can reflect a consistent trait (Réale et al. 2007; Bell, Hankison, and Laskowski 2009). Repeatability is conventionally estimated as a single-measure intraclass correlation from variance partitioning, with well-established methods for point estimation and uncertainty (Lessells and Boag 1987; Nakagawa and Schielzeth 2010). A large meta-analysis across taxa places the mean laboratory repeatability of behavior at roughly $R = 0.37$ (Bell, Hankison, and Laskowski 2009), and reproducibility has been an explicit concern in zebrafish neuroscience specifically (Gerlai 2019).

Two gaps motivate this reanalysis. First, the evidence on the repeatability of zebrafish novel-tank behavior is almost entirely *long-interval*: reported values span $R \approx 0.10$–0.59 over weeks to months (Baker et al. 2018; Thomson et al. 2020; Johnson et al. 2025; Oswald, Singer, and Robison 2013), and short-term reliability — across multiple sessions within a few days — is largely unexamined for adults (the closest evidence is in larvae, where locomotor patterns become consistent across days and morning-to-afternoon sessions (Fitzgerald et al. 2019)). Yet several studies find the first test session behaves differently from later ones: Thomson et al. (2020) reported week-1-to-2 repeatability ($R = 0.25$) far below subsequent intervals and attributed it to fish needing one tank experience before stable behavior emerges. A design with several sessions over a few days can characterize this short-term structure directly —

how quickly behavior stabilizes, and how reliable it is session-to-session, within-day, and day-to-day. Second, novel-tank reliability is reported inconsistently: studies variously give a raw test–retest correlation, a single-measure repeatability, or an internal-consistency coefficient (Cronbach's α) over several sessions, and these are not on the same scale, which makes cross-study comparison and meta-analysis difficult (Parker et al. 2016).

Here we reanalyze a dataset well suited to both gaps: 103 adult zebrafish, each tested in six novel-tank sessions over three days (two per day, morning and afternoon). The six repeated measures permit a clean separation of the novelty response from familiar-tank behavior and a properly variance-partitioned estimate of short-term repeatability. We are explicit that this is *short-term* repeatability of among-individual differences over a three-day window: it is necessary but not sufficient for a stable personality trait, and over so dense a schedule the stable component it captures may partly reflect persistent post-handling or physiological state rather than lifelong personality. The question we can answer is how consistently fish differ from one another over days, and how quickly that consistency emerges — not whether the differences persist for life. We reanalyzed the data in a newly built Python pipeline that re-implements our laboratory's earlier Mathematica analysis from the same distortion-corrected tracking output; this is a within-laboratory reanalysis, not an independent replication, and the convergence check below tests only that the new pipeline reproduces the original summary statistics before we extend them. One feature of the original tracking bounds interpretation throughout and is stated at the outset: it is frame-differencing and registers a fish only when it moves, so periods of freezing — a primary anxiety response — are not directly observed; missing positions are interpolated, which recovers within-session bottom freezes but not freezing before a session's first detection, and the activity measures are movement-conditional (Methods). Within these constraints we pursue four aims: (i) to confirm that the new pipeline reproduces the original reliability estimate; (ii) to re-express that reliability as a single-measure repeatability that can be benchmarked against the animal-personality literature, and to quantify the contribution of batch; (iii) to determine whether the first exposure measures the same construct as later sessions, and to characterize the within-day and day-to-day structure of short-term reliability; and (iv) to compare the reliability of pct_top with that of other commonly reported novel-tank measures. (The dataset includes two strains, but every testing batch was strain-pure, so strain is completely confounded with batch and is not analyzed here; see Methods.)

2 Methods

2.1 Strain is not analyzed (confounded with batch)

The dataset contains two strains (5G inbred, AB wild type), but each of the 13 testing batches was strain-pure (5G: batches 1, 3, 5, 7, 9, 12; AB: 2, 4, 6, 8, 10, 11, 13; verified by `src/11_qc_checks.py`). There is no within-batch strain variation, so strain is completely confounded with batch and with everything that varied between batches — testing date (January–March 2014), room and water conditions, handler, and per-batch distortion-correction calibration. A "5G vs AB" contrast would be identical to "the batches that ran 5G vs those that ran AB" and could not be attributed to genotype.

We therefore do not analyze strain. The batch variance component reported in the reliability model below is a separate, nuisance term and does not bear on this point.

2.2 Fish Identity Across Sessions

2.2.1 Physical tank randomization

Each batch consisted of 8 fish tested simultaneously in an 8-tank apparatus over 3 consecutive days (2 sessions per day, AM and PM), yielding 6 sessions per fish. Within each session, fish were assigned to physical tank positions 1–8. **The assignment was re-randomized for each session** so that no fish occupied the same physical tank position across all 6 runs. This counterbalances any tank-specific effects (position in the camera frame, lighting gradients, minor water quality differences) on pct_top estimates.

The tank-to-fish assignment for every session of every batch is recorded in the tank randomization decode workbook:

`p01bruno_r26/data/source/tank_randomization_decode.csv` (derived from `P1-Tank Randomization Decoding v2.xlsx`)

This workbook lists, for each session (identified by video filename and date), which physical tank position (1–8) contained which fish (identified by fish_within_batch number 1–8). It is the authoritative source for linking tracking data — which records position by physical tank slot — to individual fish identities across sessions.

In the pipeline, `src/01_ingest.py` reads fish identity directly from the `fishid` column of the distortion-corrected source file, which was populated by Bruno's Mathematica pipeline using this same decode workbook. The workbook is not re-read at runtime except to extract strain information; fish identity is already encoded in the source data.

2.2.2 Known exception: batch 5, runs 1 and 2

Batch 5 is the only batch where runs 1 and 2 (Day 1 AM and Day 1 PM) used **identical** physical tank positions for all 8 fish. Every other batch re-randomized between the AM and PM sessions on Day 1. This is confirmed in both the decode workbook (configurations F and G for batch 5 runs 1–2 list the same tank-to-fish mapping) and the source tracking data. The reason for this exception is not documented. See `manuscript/analysis.md` Section 9 for a discussion of the fish 56/57 anomaly in batch 5 that may or may not be related.

2.3 Session Exclusions

2.3.1 Source of exclusion flags

The pipeline inherits all session-level quality-control decisions from the original Mathematica analysis. The `include` column in `data/source/tracking_aggregate_uncorrected.csv` (originally `Z-P1-Data-Merged Raw No Interpolation-v5 aggregate outcomes.csv` from

Attempt 02) marks 35 sessions as `include=0`. All 35 are dropped before any feature extraction. See `src/01_ingest.py → load_exclude_set()`.

2.3.2 Breakdown of the 35 excluded sessions

Category	N sessions	Details
Missing video / NaN data	7	Batches 11–13; video files not recovered; NaN in source
Empty tank (no fish)	6	fish_id 138, all 6 runs — batch 13
Sick fish	4	fish_id 116, 4 of 6 runs — batch 11
Tracking failure	18	All remaining `include=0` sessions

fish_id 116 (batch 11, fish #6): Was sick. Excluded from 4 of 6 runs. Shows pct_top=1.000 in multiple runs — artifact of illness, not exploration behavior.

fish_id 138 (batch 13, fish #8): Tank was empty. No fish in the water. All 6 sessions excluded; most have 0–3 detections or NaN.

2.3.3 Why the 18 "tracking failure" sessions are not reinstated

All 18 sessions flagged `include=0` that have any tracking data were examined against the distribution of detection counts in included sessions:

- Median included session: **5,422 detections** over the 20-minute window
- 10th percentile of included sessions: **2,481 detections**

Every one of the 18 sessions falls below that 10th percentile, with most far below it:

fish_id	batch	run	n_det	note
13	1	1	89	< 2% of median
17	1	1	365	7% of median
15	1	1	450	8% of median
18	1	1	59	< 2% of median
16	1	3	302	6% of median
16	1	4	50	< 2% of median
18	1	4	109	2% of median
37	3	1	123	2% of median
38	3	5	109	2% of median
35	3	5	367	7% of median
37	3	6	227	4% of median
71	7	2	196	4% of median
84	8	1	132	2% of median
86	8	1	172	3% of median

fish_id	batch	run	n_det	note
83	8	4	73	1% of median
88	8	5	295	5% of median
86	8	5	349	6% of median
113	11	4	115	2% of median
126	12	1	142	3% of median
123	12	4	1	clearly bad

Detection counts this low mean the tracker was essentially non-functional for those sessions. Any pct_top values derived from 50–450 frames out of an expected ~5,000 are noise, not behavior. Several sessions' pct_top values superficially fall within the normal range (0.3–0.7), but that is not grounds for reinstatement — a nearly-failed tracker will sometimes return plausible-looking summary statistics by chance.

The batch 1 / run 1 spatial pattern is additional evidence of equipment/physical failure rather than fish behavior: tank positions 1, 2, 5, 6 (the top row of each column in that setup) were all excluded with detection counts of 59–450, while tank positions 3, 4, 7, 8 from the same recording session had normal detection counts. This is consistent with a camera angle or lighting problem affecting the top-row tanks in that specific session.

2.3.4 Summary

The original researcher's `include=0` QC flags are conservative and well-calibrated. No excluded sessions are reinstated in this reanalysis. The exclusion criterion is tracking quality (insufficient detections), not fish behavior, which means the exclusions are not expected to introduce bias in pct_top estimates.

2.4 Behavioral tracking and the freezing limitation

This analysis begins from the distortion-corrected coordinate output of the original Mathematica pipeline; the raw video files were not available for re-tracking (several batches' videos were never recovered). The reanalysis therefore tests the reproducibility of the *analysis* applied to that coordinate stream, not of the upstream tracking, and it inherits the tracking algorithm's biases. This bounds every interpretation below and is stated here, rather than only in the Discussion, because it changes what the primary measure represents.

The tracker is **frame-differencing**: it registers a fish only when the fish moves between consecutive sampled frames, so a completely still (frozen) fish produces no detection (NaN). Freezing — prolonged immobility, typically near the bottom — is a primary anxiety response, so how these gaps are handled determines what the measure represents. Missing positions are filled by **linear interpolation between the detections that bracket each gap**, applied to *interior* gaps only and with **no maximum gap length**; leading and trailing gaps (before the first or after the last detection of a session) have no flanking anchor and are left undefined. **pct_top is**

then computed over every frame with an interpolated position (detected plus interior-interpolated frames); the detected-frame-only measures — latency, transition count, velocity — are computed without interpolation.

This makes the freezing blind spot narrower than “freezing is invisible” would imply, but not absent, and the distinction matters for what each measure represents:

- A *within-session* freeze is bracketed by the detections immediately before and after it and is interpolated across — a fish that moves at minute 2, freezes on the bottom, and moves again at minute 18 has those 16 minutes filled by a straight line between the two anchors (there is no gap-length cutoff). Because a bottom freeze is bracketed by bottom detections, the interpolated track stays in the bottom zone and the freeze is **correctly counted as bottom-dwelling** — the only quantity pct_top needs is the top/bottom classification, which is robust whenever both anchors share a zone. Long mid-water gaps, where the straight-line assumption is weakest, are uncommon because vertical movement itself generates detections.
- The residual bias in pct_top is therefore confined to **leading/trailing freezes** — most relevant, a fish that dives and freezes on the bottom at the very start of the window, before its first detection — which are excluded and bias that session’s pct_top slightly *upward* (understating early anxiety).
- The **detected-only measures** (latency, transitions, velocity) do not interpolate and so genuinely understate behavior during immobility; they should be read as movement-conditional.

In sum, pct_top is a reasonably faithful index of vertical position that is robust to interior freezing, with a small residual upward bias only for unbracketed start/end freezes; the activity measures are movement-conditional. None of this is recoverable to the absolute time-at-location, because freezing cannot be distinguished from genuine tracking dropout, but for the reliability and individual-differences claims that are this paper’s focus a bias that is stable within an individual preserves the rank-ordering of fish.

2.5 Behavioral measures and missing-data coding

Five session-level measures are analyzed: top-half occupancy (pct_top), latency to first top entry, top↔bottom transition count, within-session lateral-position SD (x_sd), and mean swimming velocity. pct_top and velocity are computed over interpolated frames (above); a freeze gap thus enters velocity as slow, constant interpolated motion. x_sd, latency, and transitions are computed over detected frames only. All inherit the movement-conditional limitation above to differing degrees.

Latency to first top entry is coded in minutes from the start of the analysis window. Sessions in which the fish never entered the top half are coded as **20.0 minutes** (the full analysis window length); they are not dropped. This affects only **2 of 589 sessions (0.3%)**, so all measures except latency itself are insensitive to the choice. Latency’s own (already low) reliability is somewhat sensitive to it, because 20 min is an extreme value relative to the typical sub-minute latency; this is noted where latency reliability is reported.

The complete-case reliability estimates (the six-session composites and the single-measure ICCs) use the 82 fish with usable data in all six runs; the single-measure ANOVA repeatabilities use all 589 sessions. Of the fish analyzed, 21 lacked complete six-run data (11 of 48 5G, 10 of 55 AB), an attrition rate that does not differ materially between the two strains.

3 Results

We reanalyzed a novel-tank dataset of 103 adult zebrafish tested six times over three days (two sessions/day; runs 1, 3, 5 = morning, runs 2, 4, 6 = afternoon). After applying the original study's 35 quality-control exclusions, 589 fish × session records remained, with 82 fish providing complete data across all six runs. The primary measure was top-half occupancy (pct_top), the proportion of detected frames spent in the upper half of the tank (higher = more exploration, less anxiety-like behavior). The pipeline began from the same distortion-corrected tracking output as our earlier Mathematica analysis, so the first step is a check that the new implementation reproduces the original result before extending it.

3.1 The new pipeline reproduces the original reliability, expressed as single-measure R

Across the six sessions, pct_top showed high internal consistency (Cronbach's α = 0.855, 95% CI [0.801, 0.899]; α = 0.874 after adjusting for a small residual frame-position gradient), reproducing our earlier Mathematica analysis's $\alpha \approx 0.88$. The new pipeline is therefore a faithful re-implementation, and the small residual gap is attributable to differences in missing-frame interpolation (the original used carry-forward of the last detected position, the new pipeline bounded linear interpolation between bracketing detections; Methods).

Because α is a six-session *composite* reliability and is not directly comparable to the single-measure repeatability R reported across the animal-personality literature, we also estimated $R = V_{among} / (V_{among} + V_{within})$ by variance partitioning (Table 1). Single-session repeatability was $R = 0.50$ (one-way ICC, complete cases; identical via one-way ANOVA on all 589 sessions and via a mixed model). Projecting this single-session R to a six-session composite by the Spearman–Brown formula returns 0.856, reproducing the observed α almost exactly: the high α is the arithmetic consequence of averaging six sessions, not an anomalously reliable single measurement. Placed against published single-measure benchmarks — nearly all from much longer intervals (Bell, Hankison, and Laskowski 2009 lab mean $R \approx 0.37$; Thomson et al. 2020 bottom-time $R = 0.39$; Baker et al. 2018 swim-speed $R = 0.40$–0.59; Johnson et al. 2025 lower-zone $r = 0.29$) — pct_top sits at the upper end of the typical range; a short, within-days interval is expected to yield higher repeatability than the weeks-to-months intervals of those studies.

A nested variance decomposition (pct_top ~ 1 + (1|batch) + (1|fish-within-batch)) attributed only 1.9% of total variance in pct_top to batch, versus 49.8% to stable among-individual differences and 48.2% to within-individual variation, so the reliability estimate is not materially inflated by batch structure (Shishis, Tsang, and Gerlai

2022).

Three robustness points bear on this estimate. The Spearman–Brown step from single-session R to a six-session composite formally assumes parallel sessions, which Run 1 violates (below); this is one reason we report the more nearly parallel runs-2–6 estimate separately, and the close match between the predicted composite (0.856) and the observed α (0.855) shows the approximation is adequate here. Conditioning on session as a fixed effect — a *consistency* repeatability that removes any systematic run-to-run mean structure — leaves the estimate essentially unchanged (R = 0.52 vs. 0.51), so temporal drift in the mean does not inflate it; and because the confidence intervals are cluster-bootstrapped over fish, they already respect the non-independence of sessions within an individual. Finally, the pooled estimate is not an artifact of mixing the two strains: computed separately, single-session repeatability was R = 0.47 (5G) and R = 0.54 (AB), bracketing the pooled 0.51, so the strains have similar reliability even though their means cannot be compared (Methods; `src/11_qc_checks.py`).

Table 1. Reliability of pct_top, all runs versus the familiar-tank sessions (runs 2–6). R is the single-measure repeatability $V_{among}/(V_{among}+V_{within})$; α is the k-session composite (Cronbach). Spearman–Brown projects the single-session R to a six-session composite. Confidence intervals are cluster-bootstrap (ICC) or analytic (ANOVA).

Estimand	Method	Value	95% CI
Composite reliability, all 6 runs	Cronbach's α (n = 82)	0.855	[0.801, 0.899]
Composite reliability, slot-adjusted	Cronbach's α	0.874	—
Single-session R, all runs	one-way ICC (n = 82)	0.498	[0.378, 0.593]
Single-session R, all runs	one-way ANOVA (589 sessions)	0.511	[0.409, 0.598]
Single-session R, runs 2–6	one-way ICC	0.586	[0.478, 0.670]
Composite reliability, runs 2–6	Cronbach's α	0.876	—
Single R → 6-session composite	Spearman–Brown	0.856	—
Batch share of total variance	nested mixed model	1.9%	—

3.2 The first exposure is distinct, and reliability strengthens after the first day

The inter-run correlation structure (Figure 1; mean inter-run r = 0.50) shows that reliability is carried by the familiar-tank sessions: every pairwise correlation among runs 2–6 (r = 0.46–0.68) exceeds the correlation between Run 1 and any later run (r = 0.22–0.47).

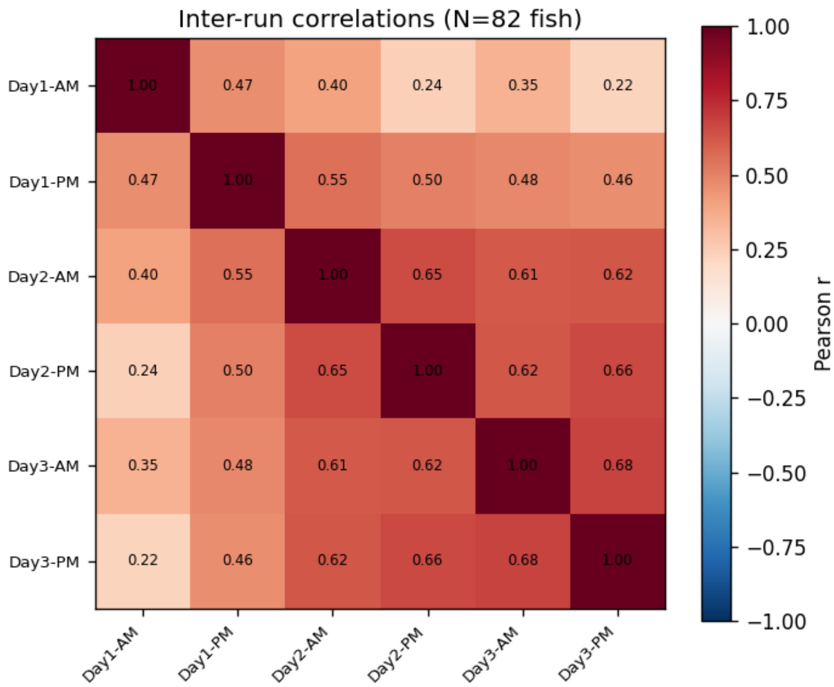

Figure 1. Pearson correlations of pct_top between all pairs of the six sessions (n = 82 fish with complete data). The first session (Day 1 AM, Run 1) correlates weakly with every other session, whereas runs 2–6 intercorrelate strongly.

Restricting the analysis to the familiar-tank sessions (runs 2–6) raised single-measure repeatability from R = 0.50 to R = 0.59 (95% CI [0.478, 0.670]) and α from 0.855 to 0.876 (Table 1). This gain was not produced by removing a population-level mean shift: the among-run mean of pct_top was essentially flat (run 1 = 0.518; runs 2–6 = 0.498), so the agreement and consistency repeatabilities were nearly identical. The flat among-run mean does not, however, imply that the assay failed to elicit a novelty response: *within* Run 1 the classic dive-then-explore trajectory is present, with pct_top rising from 0.42 in the first minute to 0.55 by minute 20, whereas Runs 2–6 are flat within-session (Figure 2) — the among-run mean is flat only because Run 1's 20-minute average coincides with the later sessions'. The Run 1 effect is instead an individual-level *reordering*: Run 1 ranks fish differently from the familiar-tank sessions (r(Run 1, mean of runs 2–6) = 0.41, versus a mean of 0.58 among runs 2–6). To avoid any appearance of selecting the higher value, we report the all-six-run R = 0.50 as the **primary** estimate and the runs-2–6 R = 0.59 as a **secondary, familiar-tank** estimate motivated *a priori* by the novelty/acclimation distinction. This pattern matches Thomson et al. (2020), who found week-1-to-2 repeatability far below later intervals and interpreted it as fish requiring one tank experience before behavior stabilizes.

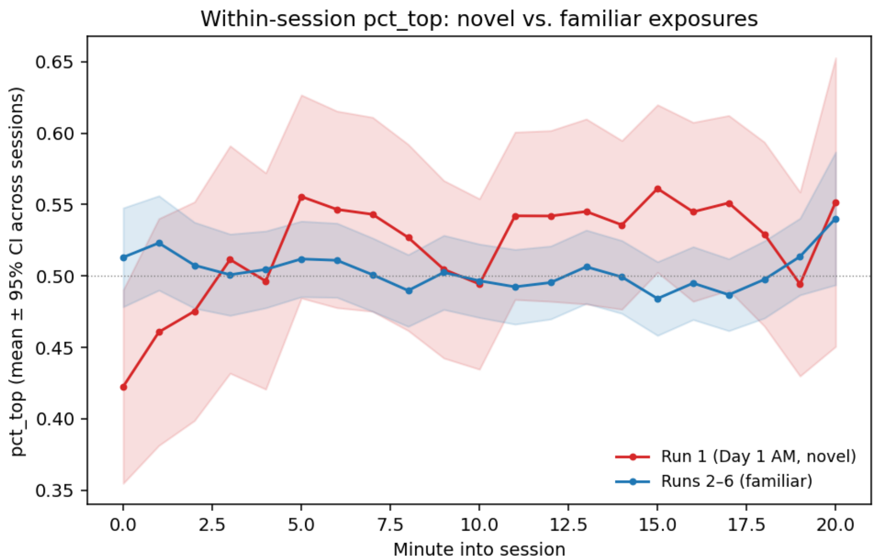

Figure 2. Within-session pct_top by minute, pooled across fish, for the novel first exposure (Run 1) versus the familiar re-exposures (Runs 2–6); bands are 95% CIs across sessions. Run 1 shows the expected dive-then-explore rise (0.42 → 0.55); the re-exposures are flat. The shallow initial dive (≈ 0.42, rather than the ≈ 0.20–0.30 of a strongly anxious cohort) indicates a modest anxiogenic response, which the freezing-blind tracker further understates (Methods).

The same first-day effect appears in the temporal structure of reliability. **Within-day** agreement between the morning and afternoon sessions rose across the three days — r = 0.47 (Day 1), 0.65 (Day 2), 0.68 (Day 3) — so a fish's two same-day measurements became more concordant once the tank was familiar. **Day-to-day** reliability across the three days, by ICC(A,1), was lower for the morning sessions (which include the Run 1 novel exposure: ICC = 0.45, 95% CI [0.32, 0.58]) than for the afternoon sessions (ICC = 0.54, [0.42, 0.66]). Both are moderate, as expected when aggregating run-to-run correlations in the 0.4–0.7 range, and both are dragged down chiefly by the first exposure rather than by a morning–afternoon difference per se. Practically, behavior in this assay is reliable from the first familiar session onward, but the very first exposure should not be pooled with it.

3.3 Reliability across the measure battery

Repeatability is not confined to pct_top. We computed four additional session-level measures — latency to first top entry, top↔bottom transition count, within-session lateral-position SD (x_sd), and mean swimming velocity — and estimated the single-session and six-session reliability of each (Table 2). Lateral movement range (x_sd) and swimming velocity were as repeatable as pct_top itself (single-session $R \approx 0.51$–0.52), establishing them as reliably repeatable over the short term; transition frequency was moderately repeatable ($R = 0.38$) and latency to first ascent least ($R = 0.33$). These measures are not redundant with pct_top — they index distinct aspects of the response (vertical position, locomotor activity, and initial latency) and differ in how reliably they do so — so a study reporting only top-half occupancy discards reliable, complementary information.

Table 2. Reliability of five novel-tank measures. R = single-session repeatability (one-way ANOVA, all sessions); α = six-session composite reliability (complete cases, n = 82). pct_top is computed over interpolated frames; latency, transitions, and velocity are movement-conditional (Methods).

Measure	R (single-session)	α (6-session)
pct_top (top-half occupancy)	0.511	0.855
x_sd (lateral movement range)	0.521	0.860
velocity (swim speed)	0.514	0.828
transitions (zone crossings)	0.376	0.764
latency (to first top entry)	0.327	0.533

3.4 Summary

This reanalysis (i) confirms our earlier reliability result with a newly written pipeline (α = 0.855 ≈ original 0.88) and re-expresses it in the field's standard single-measure currency (primary single-session $R \approx 0.50$; familiar-tank sensitivity estimate $R \approx 0.59$), showing via Spearman–Brown that the high composite α and the single-session R are the same fact on different scales; (ii) shows the first, novel exposure measures a distinct, weakly reproducible response, with within-day and day-to-day reliability both strengthening after the first day; and (iii) finds that lateral movement and swimming speed are as reliable over the short term as the conventional pct_top measure, while transitions and especially latency are less so.

4 Discussion

We reanalyzed a six-session zebrafish novel-tank dataset to characterize the short-term reliability of its behavioral measures — a timescale (multiple sessions within days) that the adult literature has largely left unexamined. Throughout, we mean by "reliability" the short-term repeatability of among-individual differences over the three-day window — a necessary but not sufficient condition for a stable personality trait, and over so dense a schedule the stable component may partly reflect a persistent post-handling or physiological state rather than lifelong personality. Three conclusions follow. First, fish differ from one another consistently over this short interval (single-measure $R \approx 0.50$), and a newly written pipeline reproduces our earlier reliability estimate. Second, the first, novel exposure measures a partly distinct construct, and reliability strengthens once the tank is familiar. Third, reliability extends beyond the conventional measure: lateral movement and swimming speed are as repeatable as top-half occupancy. We draw out the reporting and design implications below.

4.1 Reliability, and the importance of reporting it on a comparable scale

The reanalysis reproduced our earlier result: Cronbach's $\alpha = 0.855$ against the original ≈ 0.88, the small gap attributable to the change in missing-frame interpolation (carry-forward to bounded linear interpolation; Methods). Because both schemes keep a bottom freeze in the bottom zone, they largely agree, consistent with the small size of the gap. This is a within-laboratory reproducibility check on the analysis, achieved by re-implementing the original Mathematica pipeline in Python — not an independent replication — and it gives confidence that the extensions below rest on a correct re-implementation.

More consequential is how the reliability is expressed. A composite α of 0.88 appears far above the field's typical repeatability — Bell, Hankison, and Laskowski (2009) report a laboratory mean $R \approx 0.37$ — but the comparison is misleading: α is the reliability of a six-session *average*, whereas the literature reports single-measure repeatability. Once placed on the same scale, pct_top showed a single-session $R \approx 0.50$, and the Spearman–Brown projection of that value to six sessions reproduces α almost exactly. The measure is reliable and, for a short within-days interval, sits at the upper end of published estimates (cf. Thomson et al. 2020, $R = 0.39$; Baker et al. 2018, $R = 0.40$–0.59; Johnson et al. 2025, $r = 0.29$), all of which used much longer intervals over which repeatability is expected to be lower. We therefore recommend that novel-tank reliability be reported as a variance-partitioned single-measure R with the number of sessions stated, rather than as a composite α or a raw test–retest correlation, so that values are comparable across studies and usable in meta-analysis (Parker et al. 2016). That batch explained under 2% of pct_top variance is a reassuring secondary point on a factor often left uncontrolled (Shishis, Tsang, and Gerlai 2022); it bears on the reliability estimate only.

4.2 The first exposure measures a different construct

Restricting the analysis to the familiar-tank sessions raised repeatability from $R \approx 0.50$ to ≈ 0.59. This gain did not come from removing a population-level habituation trend — the among-run mean was essentially flat, so agreement and consistency repeatabilities were nearly identical — but from the first exposure ranking individuals differently from every later session. The novel exposure is thus not merely noisier; it measures a partly distinct construct, the response to genuine novelty, before stable familiar-environment behavior emerges. The same pattern runs through the temporal structure: within-day morning-to-afternoon agreement and day-to-day reliability both strengthened after the first day. This mirrors Thomson et al. (2020), who found week-1-to-2 repeatability far below later intervals, and the larval evidence that intra-individual consistency emerges with repeated testing (Fitzgerald et al. 2019).

The practical implication is direct. Many novel-tank studies test each fish once, and if that single trial is the first exposure it will capture the novelty response with comparatively weak short-term repeatability. Designs intended to estimate stable individual differences should either include an acclimation session that is excluded from analysis, or model the first exposure separately rather than pooling it with subsequent sessions.

4.3 Reliability beyond top-half occupancy

Lateral movement range and swimming velocity were as repeatable as pct_top, and transition frequency moderately so, establishing them as reliably repeatable over the short term rather than incidental locomotor noise; only latency to first ascent was weakly repeatable. Because these measures carry reliable information about distinct aspects of the response — locomotor activity and initial latency, alongside vertical position — and are not redundant with top-half occupancy (Blaser and Rosemberg 2012), we recommend reporting at least one activity-axis measure (transition count or velocity) alongside pct_top. An analysis confined to top-half occupancy discards a reliable and complementary dimension of the behavior.

4.4 Limitations

Several limitations bound these conclusions. The first, foregrounded in the Methods because it shapes the measures, is the tracker: it is frame-differencing and registers a fish only when it moves, so freezing — a core anxiety behavior — is indistinguishable from tracking dropout (Egan et al. 2009; Cachat et al. 2010). Interpolation between bracketing detections recovers *within-session* bottom freezes (counted as bottom-dwelling), so pct_top is more robust to freezing than a "detected-frames-only" measure would be; the residual bias is confined to freezes before a session's first detection, which mildly inflate pct_top and understate early anxiety, while the movement-conditional measures (latency, transitions, velocity) understate immobility more directly. We have since adopted tracking systems that do not share this blind spot; re-examining these same questions under freezing-aware tracking is a natural next step, and would also quantify how much the frame-differencing limitation distorts each measure. Sex was not recorded and could not be modeled, despite consistent sex effects on zebrafish behavior and its repeatability (Philpott et al. 2012; Thomson et al. 2020; Johnson et al. 2025); because sex is a stable between-individual attribute, sex differences load onto among-individual variance and are folded into our repeatability estimates. The estimates derive from a single cohort at one facility and a short, three-day window, so they speak to short-term reliability only and require replication before being generalized across laboratories or to longer intervals (Gerlai 2019). The non-ascender latency coding (20 min) affects only 2 of 589 sessions, so all measures except latency's own reliability are insensitive to it; two fish showed an unexplained simultaneous behavioral reversal and were retained as-is. Finally, the dataset contains two strains, but every testing batch was strain-pure, so strain is completely confounded with batch (testing date, handling, per-batch calibration) and no strain comparison is possible; we therefore make none.

4.5 Conclusion

Over a short, within-days interval, individual differences in top-half occupancy in the novel-tank test are reliably repeatable, with a single-measure repeatability of about 0.50 that the composite α restates on a six-session scale. The first exposure is a partial exception — it measures the novelty response and should be treated as acclimation — and reliability stabilizes thereafter, within-day and across days. Reliability is not unique to top-half occupancy: lateral movement and swimming speed are equally repeatable over this interval. Reporting reliability as a single-measure R with the number of sessions stated, separating the first exposure from familiar-tank behavior, and measuring more than top-half occupancy would together yield a more reliable and more complete characterization of behavior in this widely used assay.

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